CHINOOK

Project Proposal

This dashboard analyzes the performance of the Chinook Digital Music Store, highlighting sales trends, customer behavior, and popular music genres across markets. It helps stakeholders understand overall performance and identify key revenue drivers.

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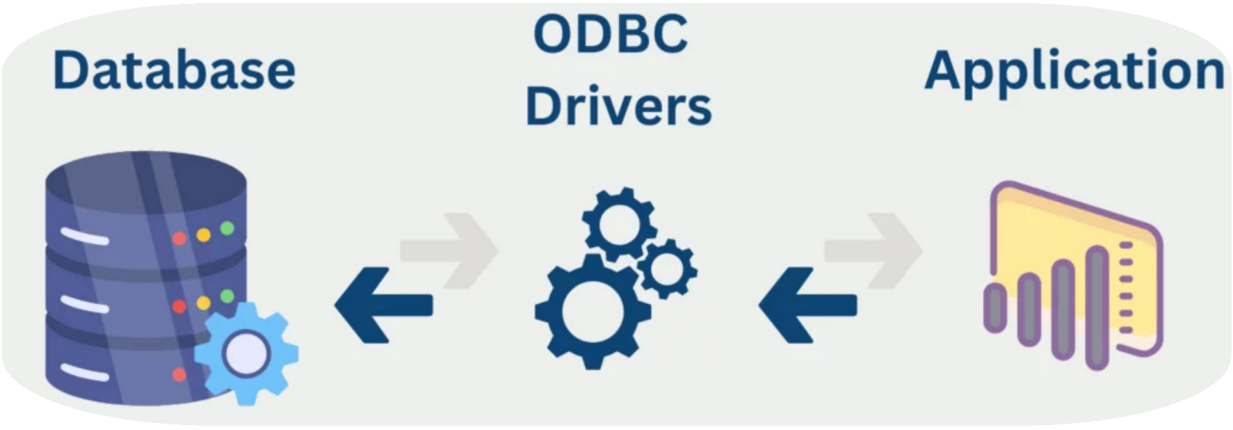
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Data Source

The data used in this dashboard comes from the Chinook SQL database, which contains information about digital music sales, customers, artists, albums, tracks, and invoices. The database simulates a digital music store that sells music tracks to customers worldwide.

The data was connected to Power BI using an ODBC (Open Database Connectivity) connection, allowing direct access to the SQL database tables. After establishing the connection, the required tables were imported into Power BI and used for further data transformation and modeling.

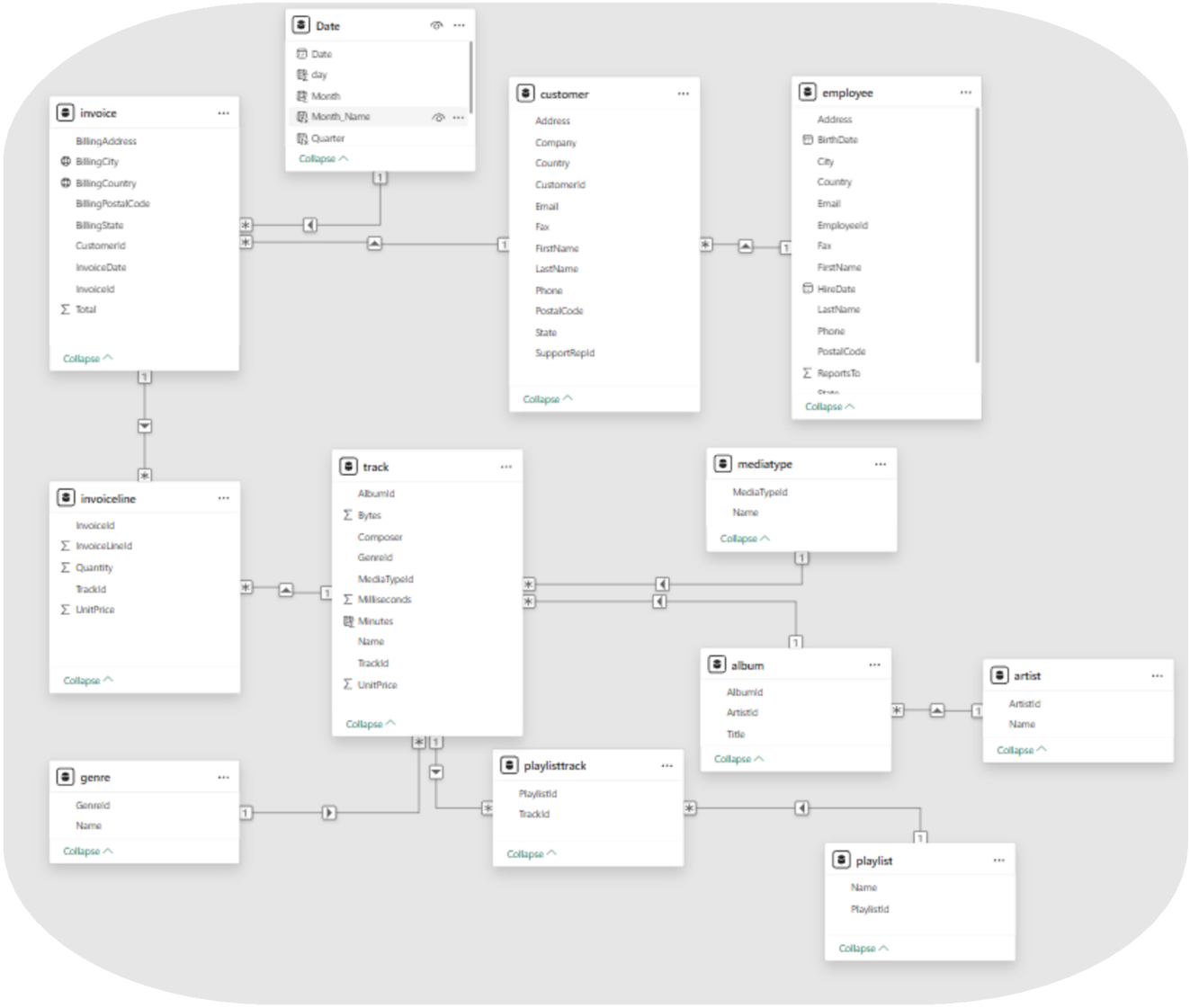
The main tables used in this analysis include customer information, invoices, invoice details, tracks, albums, artists, genres, and employees, which together enable analysis of sales performance, customer behavior, and music preferences.



Data Model

The data model was built in Power BI by establishing relationships between the main tables from the Chinook database to enable efficient analysis and filtering across the dashboard. The model follows a structured approach where sales-related tables are connected with descriptive tables such as customers, tracks, artists, and genres.

Date table was created and integrated into the model to properly manage time-based analysis and support accurate reporting for dates, months, season and years. The complete data model is illustrated in the figure below.



Key Metrics & Calculations

Date Table Calculations

1. **Date**
Definition: A calendar table generated to cover the date range used in the analysis.

Purpose: Provides a dedicated time dimension for accurate time-based reporting.

2. **Month**
Definition: Numeric representation of the month extracted from the date column.

Purpose: Used for sorting and performing month-based analysis.

3. **Month Name**
Definition: Text representation of the month (e.g., Jan, Feb).

Purpose: Improves readability in visualizations and reports.

4. **Year**
Definition: Extracted year from the date column.

Purpose: Enables yearly comparisons and trend analysis.

5. **Quarter**
Definition: Quarter of the year derived from the date column (Q1–Q4).

Purpose: Supports quarterly performance analysis.

6. **Season**
Definition: A calculated column that categorizes months into seasons (Winter, Spring, Summer, Autumn).

Purpose: Allows analysis of sales patterns based on seasonal trends.

DAX Measures

1

Total Tracks Sold

Definition: Sum of the quantity of tracks sold from the invoice line table.

```
1  Total Tracks Sold = SUM(invoiceline[Quantity])
```

Purpose: Measures the total number of music tracks purchased by customers across all transactions.

2

Total Tracks

Definition: Distinct count of unique track IDs available in the dataset.

```
1  Total Tracks = DISTINCTCOUNT(track[TrackId])
```

Purpose: Indicates the total number of different music tracks offered in the store catalog.

3

Total Sales

Definition: Total revenue calculated by multiplying unit price by quantity for each sold track.

```
1  Total Sales = SUMX(invoiceline, invoiceline[UnitPrice] * invoiceline[Quantity])
```

Purpose: Represents the overall revenue generated from music track sales.

4

Total Artists

Definition: Distinct count of artist IDs in the dataset.

```
1 Total Artists = DISTINCTCOUNT(artist[ArtistId])
```

Purpose: Represents the total number of artists whose music is available in the store.

5

Total Invoices

Definition: Count of all invoice records in the dataset.

```
1 Total Invoices = COUNT(Invoice[InvoiceId])
```

Purpose: Shows the total number of purchase transactions made by customers.

6

Total Customers

Definition: Distinct count of customer IDs.

```
1 Total Customers = DISTINCTCOUNT(customer[CustomerId])
```

Purpose: Measures the total number of unique customers who made purchases.

7

Total Albums

Definition: Distinct count of album IDs.

```
1 Total Albums = DISTINCTCOUNT(album[AlbumId])
```

Purpose: Indicates the number of albums.

8

Customer Countries

Definition: Distinct count of countries where customers are located.

```
1 Customer Countries = DISTINCTCOUNT(customer[Country])
```

Purpose: Helps analyze the geographical distribution of the customer base.

9

Average Order Value (AOV)

Definition: Average value of the total amount per invoice.

```
1 Avg Order Value = AVERAGE(invoice[Total])
```

Purpose: Measures the typical spending amount per customer transaction.

10

Active Customers

Definition: Distinct count of customers who made purchases during the previous year.

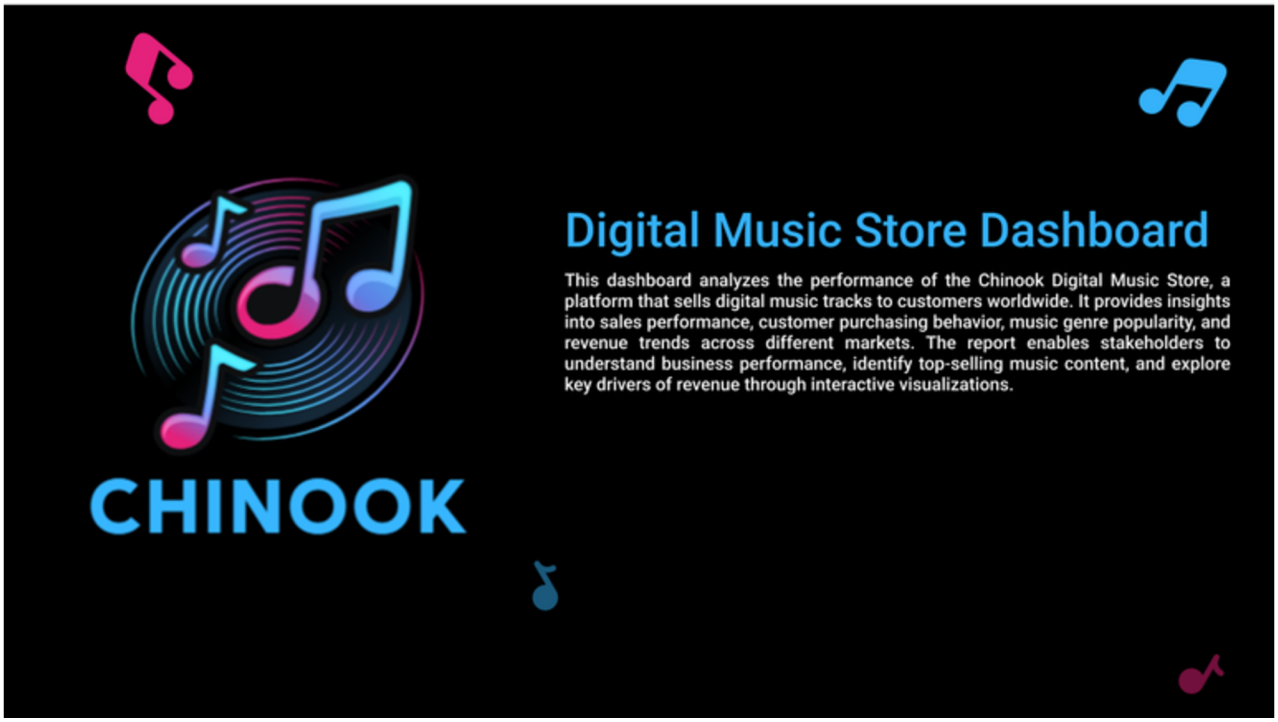
```
1 Active Customers = CALCULATE(DISTINCTCOUNT(Customer[CustomerId]),FILTER(Invoice,YEAR(Invoice[InvoiceDate]) = YEAR(TODAY
  ()) - 1))
```

Purpose: elps identify recently engaged customers and track customer activity trends.

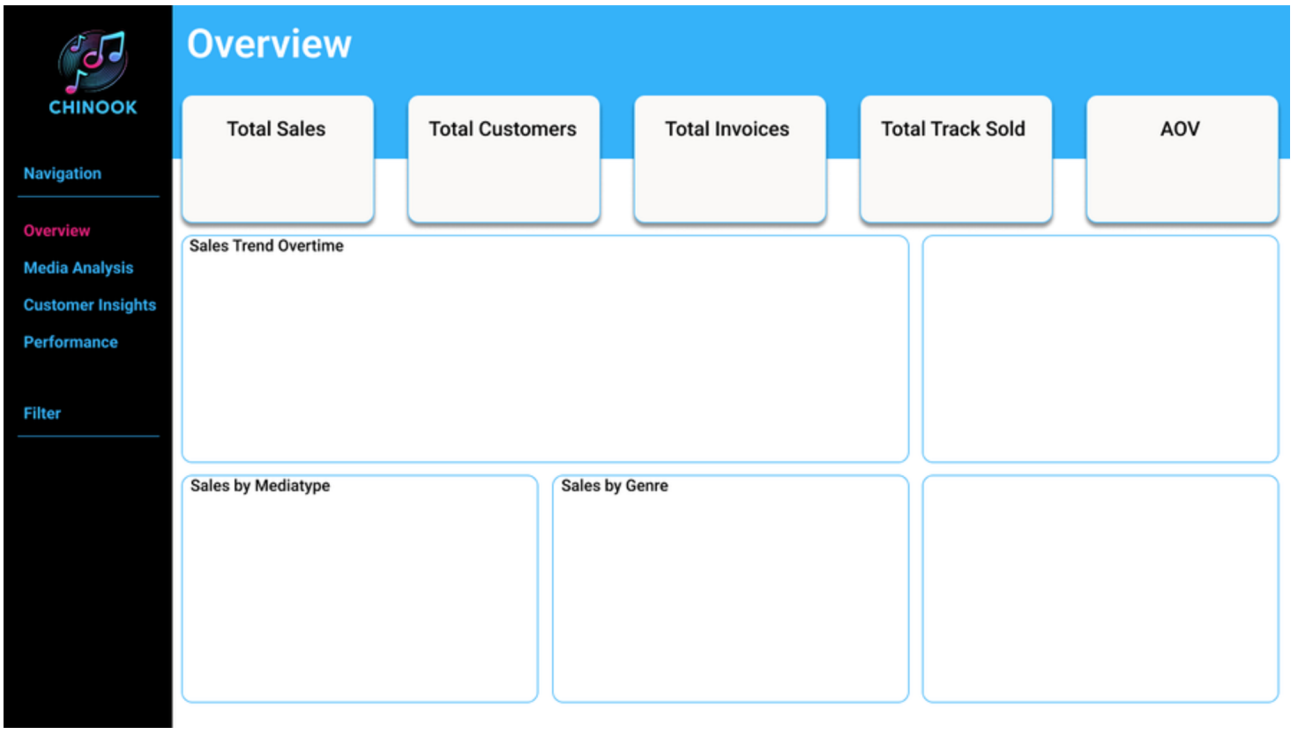
Dashboard Structure

The dashboard layout and design were first planned and structured using Figma to create a clear and user-friendly interface before implementation in Microsoft Power BI. This step helped organize the placement of KPIs, charts, and filters to ensure an intuitive flow for users when exploring the data.

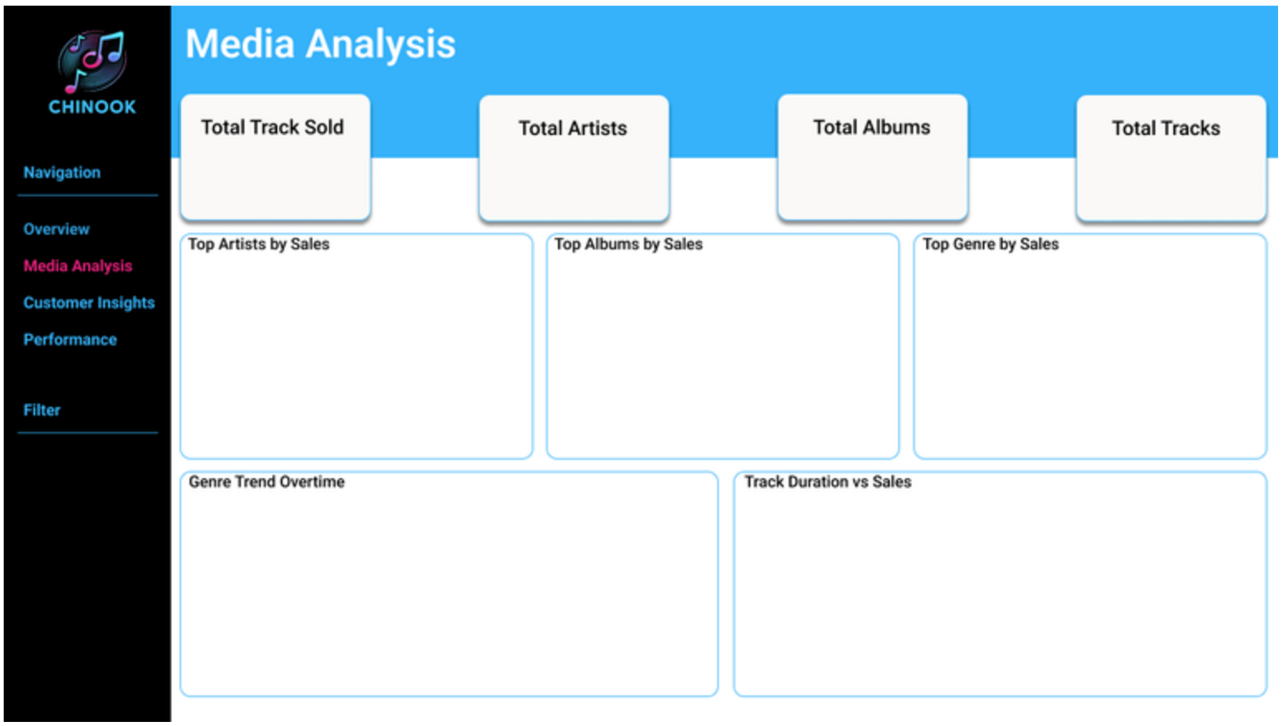
1 Home Page



2 Overview Page

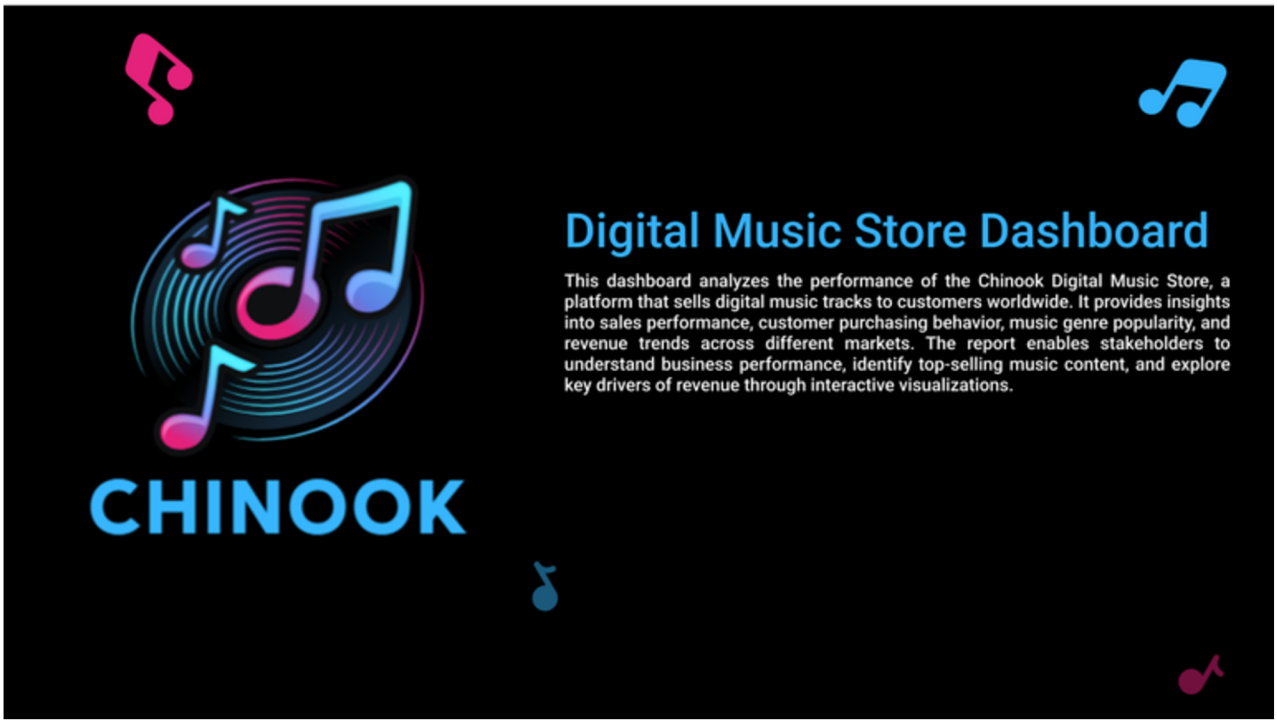


3 Media Analysis Page



Final Dashboard

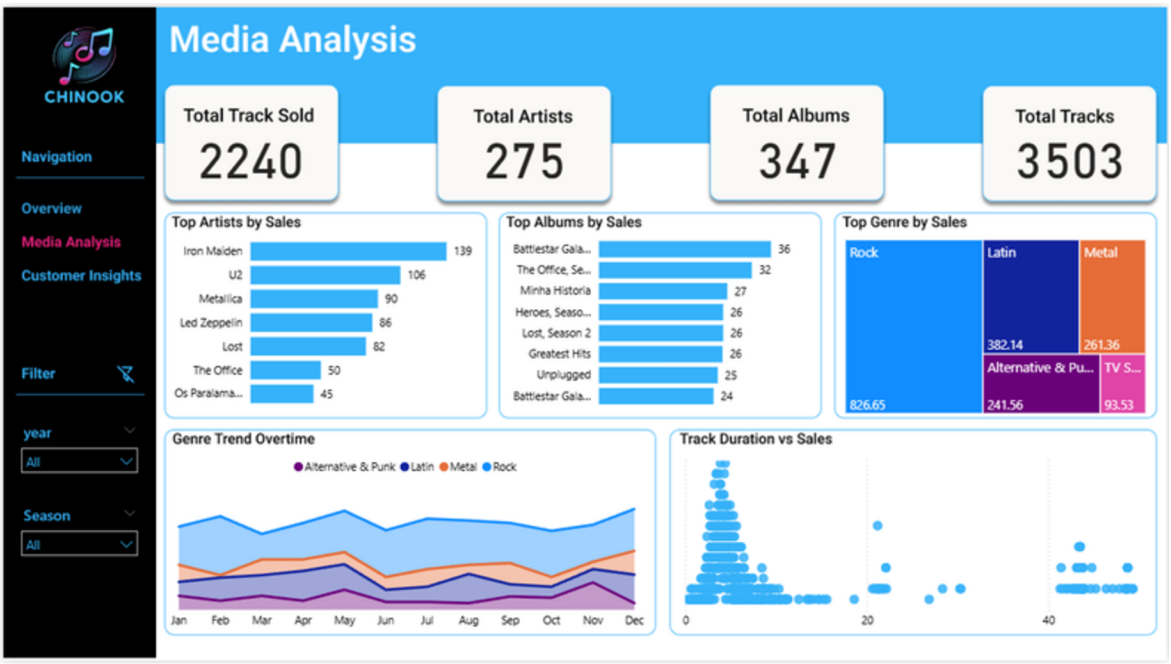
1 Home Page



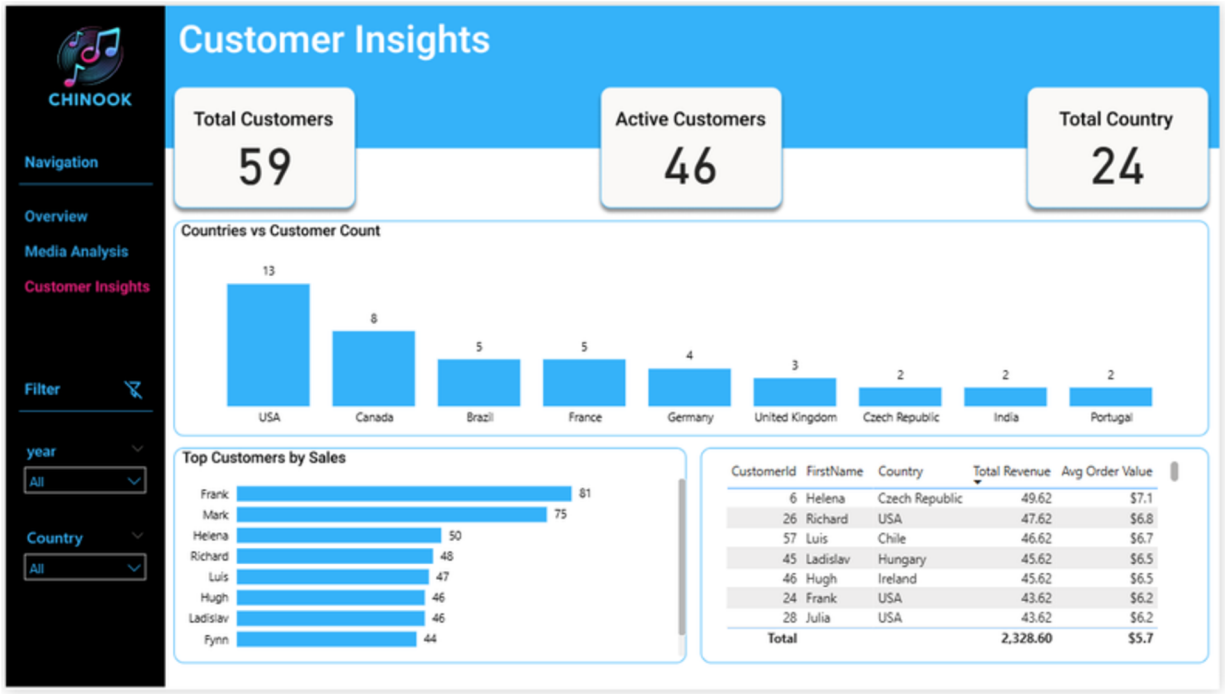
2 Overview Page



3 Media Analysis Page



4 Customer Insights Page



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